

District Energy Digital Twin

DeltaV Simulation · HMI Development · Control Narrative

OBJECTIVE

Build a control system and digital twin of the Senákw District Energy System (DES) in Emerson DeltaV Distributed Control System (DCS) - a 32 MW heating / 15 MW cooling plant serving Canada’s largest Indigenous development, enabling control verification, PID tuning, and operator training before physical commissioning.

MY CONTRIBUTION

Simulation modules: built first-principles DeltaV composite blocks for heat exchangers, pumps, valves, and energy transfer stations (ETS), etc. simulating flow, pressure, and temperature behaviour

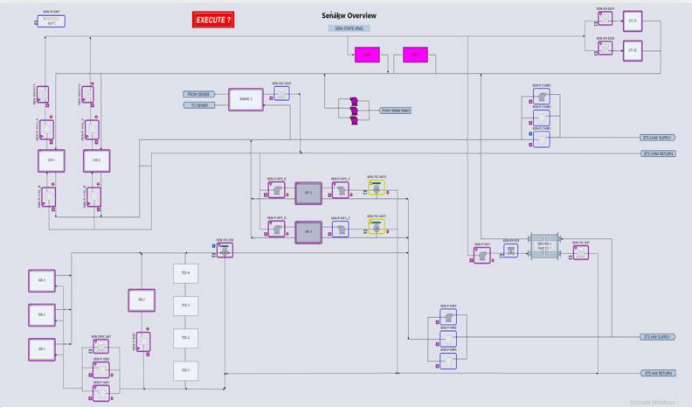
HMI ownership (DeltaV Live): designed all operator graphics to ISA-101 standard (L1–L4 hierarchy); custom GEMs for each equipment class

Control narrative reference: studied the control narrative to verify the control logic was correctly implemented, and to identify any discrepancies requiring design changes

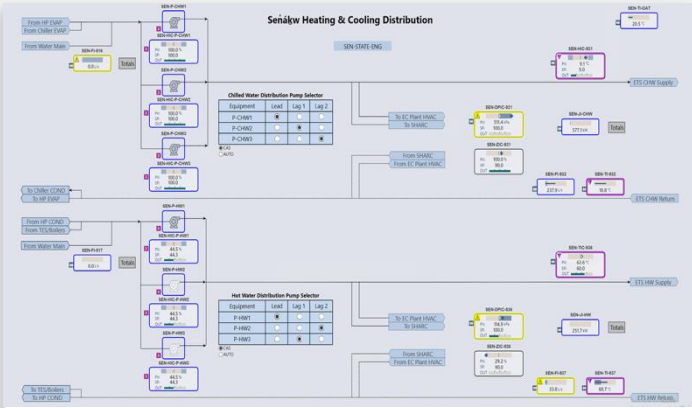
Control design validation: identified and resolved HP circulation pump issue via simulation; variable speed fix implemented to eliminate chiller recirculation inefficiency

OUTCOME

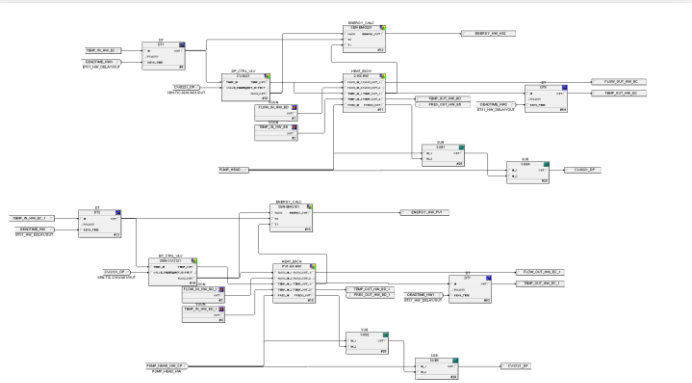
Digital twin operational for all 3 buildings and full energy center. Simulation revealed control design flaw corrected pre-commissioning. HMI delivered covering L1–L4 with ISA 101-compliant grey graphics.



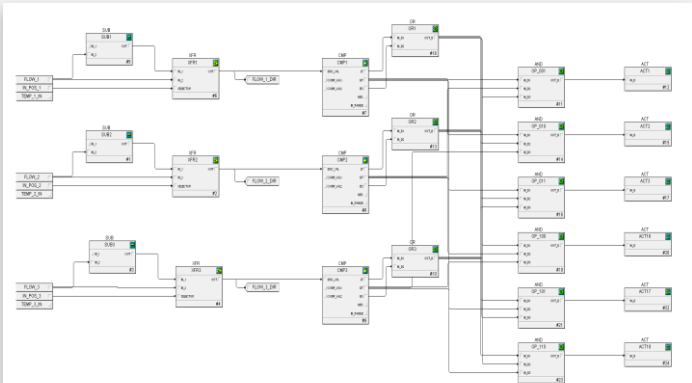
DeltaV Senákw Overview: L1 HMI showing full plant



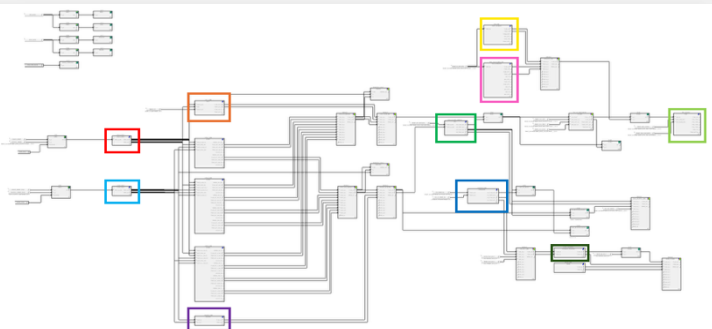
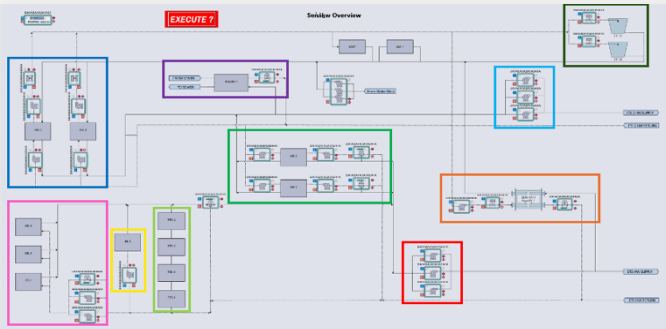
Heating & Cooling Distribution: L3 detail with pump selectors & PID faceplate



Simulation: ETS composite module internals (heat exchanger + valve)



Simulation: 3-way valve junction temperature engine, 6-state logic



DeltaV simulation module overview (left) and corresponding block diagram location (right)